

1 Analysis of the given model

The given model was trained using the following hyperparameters: $epochs = 10$, $batch\ size = 128$, $learning\ rate = 0.001$ and $optimizer = Adam$. Loss figure shows an unstable training because the training loss curve is extremely steep, this problem could be solved by decreasing the learning rate value. Furthermore, training and validation curve are still decreasing, the epochs could be increased and the performance will improve.

2 Optimization

The first optimization step was to reduce the learning rate to 0.0001 and increase the number of epochs to observe better the behaviour of the model. An early stopping technique was also used to avoid overfitting the patiente was setted to 5 and the min_delta parameter was setted to 0.02. This was done to reduce overfitting and because with $min_delta = 0$ the callback did not stop the training even though the loss curves stop decreasing.

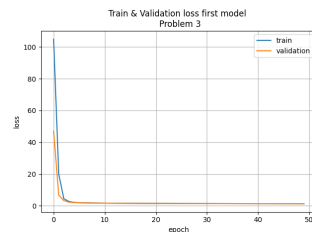


Figure 5: Loss of the first optimized model

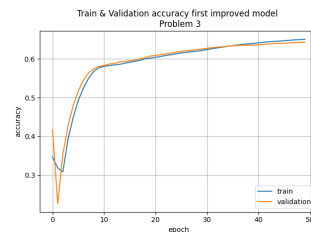


Figure 6: Accuracy of the first optimized model

Figures 5 and 6 show the results obtained with the new hyperparameter. There is a bit of overfitting around and after the thirtieth epoch. Moreover, after analyzing the weight heatmap, it was found that the weights in each layer had average values close to 0, meaning they had a sparse representation. Because of this, using L1 regularization didn't make much sense, instead it was replaced with L2 regularization which promote a less sparse weight distribution and penalize severely large weights. After changing the regularization techniques, a Dropout layer with a rate of 0.25 was added between the input layer and the first hidden layer. This improve the model's generalization and reduce overfitting.

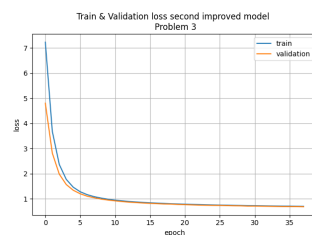


Figure 7: Loss of the second optimized model

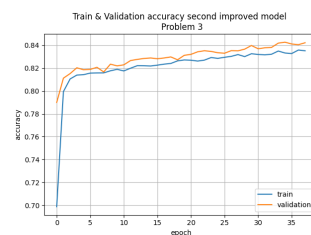


Figure 8: Accuracy of the second optimized model

Figures 7 and 8 show good results. There is no overfitting, and the model demonstrates good generalization with a smooth loss curve. The changes introduced in the hyperparameters and the applied regularization techniques have improved the model, ensuring good performance.